

# **SOFTWARE ENGINEERING PROJECT**

**TOPIC:**

**DISASTER MANAGEMENT AND RELIEF SYSTEM**

**SUBMITTED TO:**

**MS. ASIA SAHAB**

**SUBMITTED BY:**

**HERMAIN KASHIF 240901026**

**SYEDA ZAINAB RAZA 240901035**

**EIRAJ FAISAL 240901038**

# **Part completed by Syeda Zainab raza, Hermain kashif, Eiraj Faisal (35, 26, 38)**

## **INTRODUCTION:**

When a disaster strikes, countless number of people suffer from the consequences. There is a large amount of damage to human resources in these cases. In scenarios like these, not receiving proper help can be a major issue for the affected victims. To solve this problem, there must be a proper connection between NGO's, volunteers, donors, and victims.

The disaster management and relief system helps solve all these problems promptly. This system helps connect all the participating NGO's, volunteers, and donors to the victims through a centralized system. This system accepts complaints and help requests from disaster victims and passes them on to individuals willing to assist. These people then accept the request and contribute where needed the most.

This system helps ensure all the activities are transparent, well-organized, and efficient. This system also helps the government to keep track of available resources to reduce confusion and malpractice during emergencies.

The main goal of this project is to efficiently provide help and relief to the people in need so they can recover fast from the damages they faced.

# **Part completed by Syeda Zainab raza (240901035)**

## **FUNCTIONAL REQUIREMENTS:**

### **1: Volunteer registration:**

- The volunteers must register themselves with their names and personal information.
- The user must add information about their role in the system.
- Every user must have their designated login ID.

### **2: Complaint management center:**

- The user can enter their complaints in the system complaint center.
- The victims can mention their needs and wants in case of any disaster situation.
- The system must receive and process the complaint and forwards it to the NGO'S and volunteers for them to take action.

### **3: Assigning tasks to volunteers:**

- The complaint of the victim is processed by the system.
- The system shows the complaint on the user's side.
- The system assigns the complaint to the available volunteers by tracking the volunteer activity.
- The volunteers who are assigned the task then take action to resolve the issue.

#### **4: Management of resources:**

- The system keeps track of the resources that are available for the victims.
- The authorized personnel can also alter the records by entering or deleting any extra information.
- They can enter the any updates in the donation about any new donations.

#### **5: Track of volunteer activities:**

- The system keeps track of the activities of different volunteers.
- It checks if the volunteers are available.
- It shows what volunteers are assigned to what task at what time.
- The volunteers can add and update their status according to their availability.

#### **6: Checking shelters and relief camps:**

- The system keeps a track of the availability of shelter camps for the victims in case of a disaster.
- The volunteers can also update the availability or unavailability of the shelters.
- The system can also keep track of the location of the relief camps
- It can also track the amount of people a relief camp can accommodate.

#### **7: Disaster alert system:**

- The system helps the volunteers can get a warning message or an alert in case of any kind of emergency situation.
- This will help the volunteers to take an immediate action upon the situation and prepare necessary reliefs for the affected people.
- The alerts are traced so the system can inform the ngo/volunteers about the location.

## **8: Generation of reports:**

- The system is able generated well written reports for the user.
- These reports may contain information about a specific disaster.
- These reports can be about the number of resources used in a specific situation or at a specific location.
- These reports help the users to keep track of the relief management they carried out for the victims.

## **9: Handling system without internet:**

- This system helps manage and helps enter the information in the system without the need of internet.
- The system and the information will synchronize as soon as the system catches any signal of internet.

## **10: Security:**

- The system must provide full confidentiality to the victims and members.
- The donors must not be able to access any information about the victims.
- Only the managers must be able to view everything about the victims, donors, NGO's etc.

# **Part completed by Eiraj Faisal (240901038)**

## **NON-FUNCTIONAL REQUIREMENTS:**

### **1: Performance requirements:**

- The system should be able to load within five seconds.
- The system maps and volunteer activity should be updated every 15 minutes to keep the system updated about their availability.
- The system should be able to handle a lot of traffic at the same time in case of any emergency.

### **2: Security Requirements:**

- The system should keep the confidential information about the volunteers, etc., restricted to the authorized people only.
- The system should also protect the information about the resource management from itself.

### **3: Usability Requirements:**

- The system should be usable, which means it should be user-friendly.
- The system is to be used by many users, such as victims, volunteers, and Donors, so the user interface should be easy to use and should not be complex so that even a layman can easily figure it out.

### **4: Scalability Requirements:**

- This system should be scalable and should be able to handle thousands of requests at a time without lagging.
- Scalability is an important factor for any system. A system that lacks it is very inefficient. So, this requirement should be fulfilled.

## **5: Maintainability Requirements:**

- The system should be maintainable and updatable.
- Any new updates should be accessible easily, and there should be room for maintenance to increase system efficiency.

## **6: Reliability Requirements:**

- The system should be reliable to its users, meaning that the victims can rely on the system and be sure that the help they need will reach them.
- Whereas Donors can rely on the system, that their donations will surely reach the needy.

## **7: Availability Requirements:**

- The system should be available to its users at all times.
- Even without internet, the users can use the system

# Part completed by Syeda Zainab raza (240901035)

## UPPER USE CASE:

### 1: Volunteer registration

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use-case body</b>    | Volunteer Registration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Trigger</b>          | When a volunteer wants to register in the system.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Pre-condition</b>    | Volunteer opens the registration page.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Basic Path</b>       | <ol style="list-style-type: none"><li>1. Volunteer selects the option 'Register as a volunteer'.</li><li>2. The system asks if the volunteer is part of an NGO or not. If yes, then the NGO name is required.</li><li>3. Then inserts personal info (name, email, etc.).</li><li>4. Volunteer selects the field he wants to volunteer in (medical, rescue operation, etc.).</li><li>5. Volunteer: add username and password.</li><li>6. The system verifies the volunteer by sending a verification message.</li><li>7. Admin approves the volunteer based on experience and history.</li><li>8. The system sends confirmation to the volunteer.</li></ol> |
| <b>Alternative Path</b> | If the system enables auto-approval, the volunteer is added to the database.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Post condition</b>   | Volunteer record is added to the database.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Actors</b>           | Volunteers, System                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Exception Paths</b>  | If the volunteer leaves any important info unentered during registration, they cannot proceed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Others</b>           | Volunteers can update profiles                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |



## 2: Complaint Management Center

|                          |                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use-case name</b>     | Complaint Management Center                                                                                                                                                                                                                                                                                                                      |
| <b>Trigger</b>           | The system helps the victims to submit help requests in case of emergencies.                                                                                                                                                                                                                                                                     |
| <b>Pre-condition</b>     | The victims must be entering their location.                                                                                                                                                                                                                                                                                                     |
| <b>Basic Path</b>        | 1: The victims enter the system and open up the complaint center.<br>2: The victims type their complaint.<br>3: The victims select the intensity of their complaint and their location.<br>4: The victims select the type of help required.<br>5: The system takes the complaint and further processes it to different NGO's and volunteers etc. |
| <b>Alternative paths</b> | If the complaint is not sent, the system tries to find a better connection to process the complaint.                                                                                                                                                                                                                                             |
| <b>Post Condition</b>    | The complaint is sent further to the system and then the system forwards the complaint to the designated admin to take action.                                                                                                                                                                                                                   |
| <b>Actors</b>            | Victims, NGO/volunteers, System                                                                                                                                                                                                                                                                                                                  |
| <b>Exception Paths</b>   | Invalid or wrong information gets rejected.                                                                                                                                                                                                                                                                                                      |
| <b>Others</b>            | The system ensures that the complaint is processed.                                                                                                                                                                                                                                                                                              |

### 3: Assigning tasks to volunteers

|                          |                                                                                                                                                                                                                                                                                                    |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use-case name</b>     | Assigning of tasks to volunteers                                                                                                                                                                                                                                                                   |
| <b>Trigger</b>           | A system handler assigns a victim request to a volunteer.                                                                                                                                                                                                                                          |
| <b>Pre-condition</b>     | There should be a victim request in the system for the system to assign to the volunteer.                                                                                                                                                                                                          |
| <b>Basic Path</b>        | 1: The system processes the victim request.<br>2: The manager assigns a victim request to the available volunteer.<br>3: The volunteer accepts the task from the system manager.<br>4: The system updates the availability of the volunteer.<br>5: The system changes the status of the complaint. |
| <b>Alternative paths</b> | If one volunteer becomes unavailable, the system manager can assign the complaint to another volunteer.                                                                                                                                                                                            |
| <b>Post Condition</b>    | The complained is marked as “complain being handled” or “complain handled successfully.”                                                                                                                                                                                                           |
| <b>Actors</b>            | System, Volunteers                                                                                                                                                                                                                                                                                 |
| <b>Exception Paths</b>   | If there is an unavailability of volunteers or an invalid complaint.                                                                                                                                                                                                                               |
| <b>Others</b>            | The system handles a victims problem successfully.                                                                                                                                                                                                                                                 |

### 4: Management of resources

|                          |                                                                                                                                                                                                                                                                  |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use-case name</b>     | Management of resources                                                                                                                                                                                                                                          |
| <b>Trigger</b>           | The system helps manage keep a track of the number of resources available for the victims collectively.                                                                                                                                                          |
| <b>Pre-condition</b>     | The NGO's and volunteers must give resources to the system.                                                                                                                                                                                                      |
| <b>Basic Path</b>        | 1: The NGO's and volunteers give their donations and enter the records into the system.<br>2: They can add and delete records from the system.<br>3: System gets updated incase of any donations from the system.<br>4: The updates appear on the system center. |
| <b>Alternative paths</b> | Resources can be donated by anonymous personnel with proper registration.                                                                                                                                                                                        |
| <b>Post Condition</b>    | Any additions or subtractions in the system are updates.                                                                                                                                                                                                         |
| <b>Actors</b>            | Donors, System, Volunteers                                                                                                                                                                                                                                       |
| <b>Exception Paths</b>   | If the system has invalid entries.                                                                                                                                                                                                                               |
| <b>Others</b>            | The system updates the records in accordance with different NGO's and volunteers.<br>NGO's and volunteers can only update their own personal records.                                                                                                            |

## 5: Track of Volunteer activities

|                          |                                                                                                                                                                                                                                                                                                              |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use-case name</b>     | Track of volunteer activities                                                                                                                                                                                                                                                                                |
| <b>Trigger</b>           | The measures and steps taken by organizations and volunteers in case of any disaster or emergency.                                                                                                                                                                                                           |
| <b>Pre-condition</b>     | The volunteer gets assigned a task.                                                                                                                                                                                                                                                                          |
| <b>Basic Path</b>        | 1: The volunteer gets assigned a task in case of any emergency by the system.<br>2: The volunteer takes action on completing the assigned task.<br>3: If the task is in process the it gives "in process".<br>4: The system updates the activity of the volunteer to "Completed" when the task is completed. |
| <b>Alternative paths</b> | If the volunteer has any problem in completing the task, the system can assign the task to any other available team.                                                                                                                                                                                         |
| <b>Post Condition</b>    | The system updates the volunteer status as the task is completed.                                                                                                                                                                                                                                            |
| <b>Actors</b>            | System, Volunteers/NGO's                                                                                                                                                                                                                                                                                     |
| <b>Exception Paths</b>   | The system is not able to update the status on time.                                                                                                                                                                                                                                                         |
| <b>Others</b>            | The system is updated only when the confirmation from both the volunteer and victim is received.                                                                                                                                                                                                             |

## 6: Checking shelters and relief camps

|                          |                                                                                                                                                                                                                                           |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use-case name</b>     | Checking shelters and relief camps                                                                                                                                                                                                        |
| <b>Trigger</b>           | Building shelters and camps for victims who loose their homes and adding the information into the system.                                                                                                                                 |
| <b>Pre-condition</b>     | The system should have the information about the availability of the shelter.                                                                                                                                                             |
| <b>Basic Path</b>        | 1: The system can add more shelters if there is a shelter made by any volunteer.<br>2: The volunteer is going to add the information about the shelter along with its location.<br>3:The system will update or add shelter in the system. |
| <b>Alternative paths</b> | The previous shelters can be reused in case there is an unavailability of shelters.                                                                                                                                                       |
| <b>Post Condition</b>    | The data will be updated according to the changes.                                                                                                                                                                                        |
| <b>Actors</b>            | System, Volunteers/NGO's                                                                                                                                                                                                                  |
| <b>Exception paths</b>   | Invalid data or wrong location provided.                                                                                                                                                                                                  |
| <b>Others</b>            | Information of the availability of shelters will be updated often.<br>The system should also provide the information about how many people can a shelter accommodate.                                                                     |

## 7: Disaster Alert system

|                          |                                                                                                                                                           |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use-case name</b>     | Disaster alert system                                                                                                                                     |
| <b>Trigger</b>           | There will be an alert or warning message in case of any disaster for the volunteers.                                                                     |
| <b>Pre-condition</b>     | There is any new complaint, news or prediction of the disaster.                                                                                           |
| <b>Basic Path</b>        | 1: The system throws a warning message in case of any threat.<br>2: The system generates an alert message for the volunteers in case of active emergency. |
| <b>Alternative paths</b> | The volunteer can reject an emergency call.                                                                                                               |
| <b>Post Condition</b>    | The messages are successfully delivered to the users of the system.                                                                                       |
| <b>Actors</b>            | System, Volunteers/NGO's                                                                                                                                  |
| <b>Exception Paths</b>   | There is a false warning or message.<br>The warning failed to process.                                                                                    |
| <b>Others</b>            | The alerts should be tracked and marked.                                                                                                                  |

## 8: Generation of reports

|                          |                                                                                                                                                                                                                                                     |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use-case name</b>     | Generation of reports                                                                                                                                                                                                                               |
| <b>Trigger</b>           | The user of the system asks for a report.                                                                                                                                                                                                           |
| <b>Pre-condition</b>     | The user must be a registered volunteer.                                                                                                                                                                                                            |
| <b>Basic Path</b>        | 1: The user asks he system to generate a report for a specific disaster event on a specified date.<br>2: The system will gather the information about the victims, resources or volunteers.<br>3: The system will generate the report for the user. |
| <b>Alternative paths</b> | Filter according to the region.                                                                                                                                                                                                                     |
| <b>Post Condition</b>    | The system generates a report and the user has a saved report.                                                                                                                                                                                      |
| <b>Actors</b>            | System                                                                                                                                                                                                                                              |
| <b>Exception Paths</b>   | Inconsistent data can cause wrong reports.                                                                                                                                                                                                          |
| <b>Others</b>            | There can be separate reports for each organization.                                                                                                                                                                                                |

## 9: Handling system offline/ without internet

|                      |                                                 |
|----------------------|-------------------------------------------------|
| <b>Use-case name</b> | System handling without internet                |
| <b>Trigger</b>       | Allows the user to enter data without internet. |

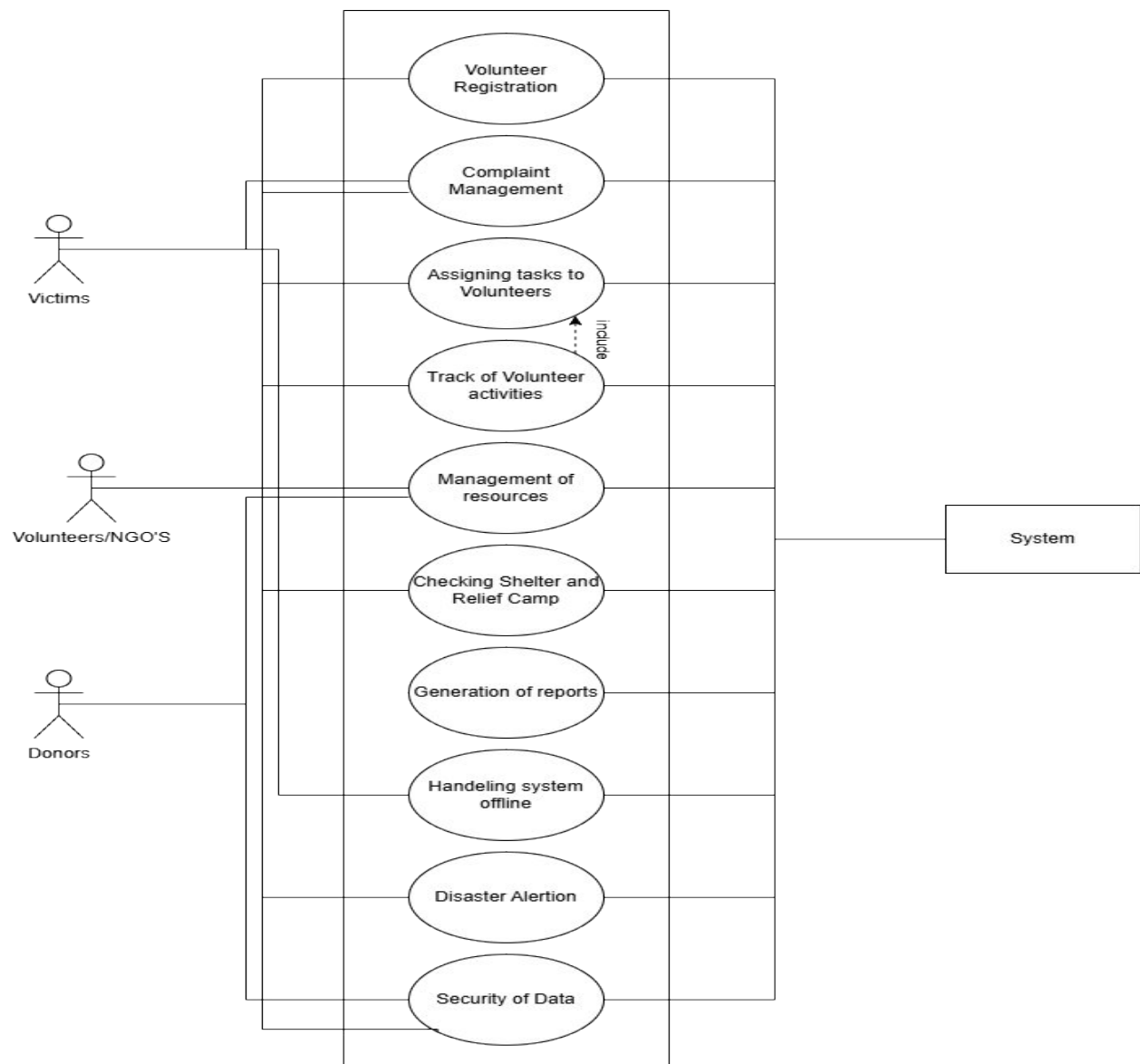
|                          |                                                                                                                                                                                                                                           |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pre-condition</b>     | The user should be able to open the system.                                                                                                                                                                                               |
| <b>Basic Path</b>        | 1: The user enters their information.<br>2: The system saves the information.<br>3: The system is not updated because of the unavailability of internet.<br>4: The system syncs the newly entered information after it gets a connection. |
| <b>Alternative paths</b> | User can trigger the synchronization manually too.                                                                                                                                                                                        |
| <b>Post Condition</b>    | The system is updated as soon as there is an availability of internet.                                                                                                                                                                    |
| <b>Actors</b>            | System, Victims                                                                                                                                                                                                                           |
| <b>Exception Paths</b>   | The file does not get in the process of sending.                                                                                                                                                                                          |
| <b>Others</b>            | The system can display the phase the file lies in during the sending process.                                                                                                                                                             |

## 10: Security and confidentiality

|                          |                                                                                                                                                                                                                                                                                                                        |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use-case name</b>     | System security and confidentiality                                                                                                                                                                                                                                                                                    |
| <b>Trigger</b>           | The user tries to login                                                                                                                                                                                                                                                                                                |
| <b>Pre-condition</b>     | The user must have a proper login info                                                                                                                                                                                                                                                                                 |
| <b>Basic Path</b>        | 1: The user enters the system.<br>2: The system asks the user to verify his identity.<br>3: The user enters his role.<br>4: The system recognizes the role.<br>5: The system asks the user for their name and assigned password.<br>6: The user can enter the system and access the information allowed by the system. |
| <b>Alternative paths</b> | The system will warn the user if the wrong password is entered.                                                                                                                                                                                                                                                        |
| <b>Post Condition</b>    | The system successfully allows the user to access their desired information.                                                                                                                                                                                                                                           |
| <b>Actors</b>            | System, Volunteers/NGO's, Donors                                                                                                                                                                                                                                                                                       |
| <b>Exception Paths</b>   | If the user doesn't know the password, they cannot enter the system.                                                                                                                                                                                                                                                   |
| <b>Others</b>            | The user can change their password by following a proper procedure in case they forget their password.                                                                                                                                                                                                                 |

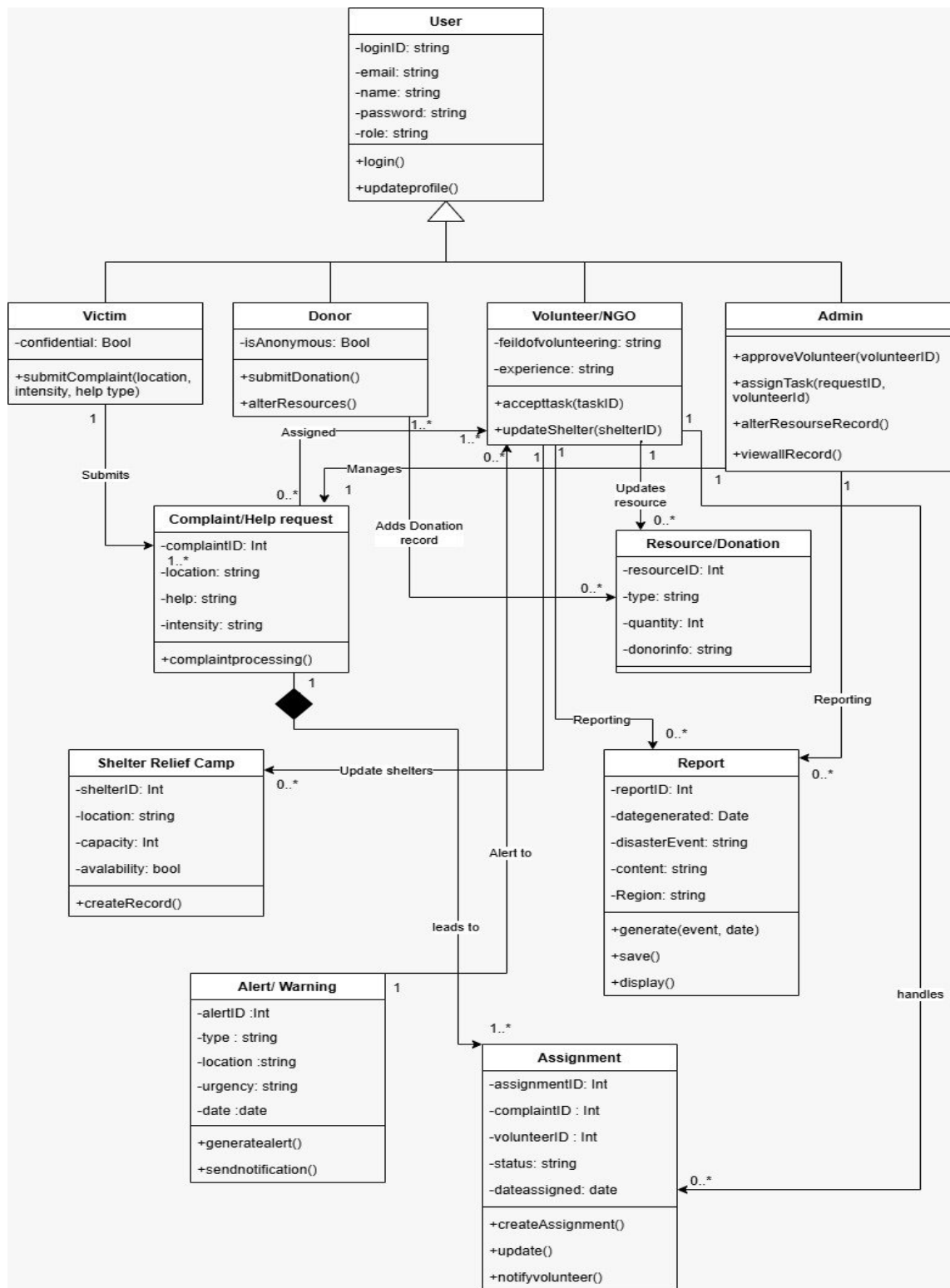
Part completed by hermain Kashif (240901026)

## UML Diagram



# CLASS DIAGRAM

- **User → Victim:** Inherits from User (**inheritance**).
- **User → Donor:** Inherits from User (**inheritance**).
- **User → Volunteer/NGO:** Inherits from User (**inheritance**).
- **Victim → Complaint:** One victim can submit many complaints (**one-to-many**).
- **Complaint → Assignment:** One complaint can generate multiple assignments (**one-to-many**).
- **Volunteer → Assignment:** One volunteer can handle many assignments (**one-to-many**).
- **Donor → Resource/Donation:** One donor can contribute multiple donations (**one-to-many**).
- **Volunteer → Resource/Donation:** Volunteers update resource records (**association**).
- **Volunteer → Shelter Relief Camp:** Volunteers update shelter information (**association**).
- **Alert → Volunteer:** One alert can notify multiple volunteers (**one-to-many**).
- **Complaint ↔ Volunteer (via Assignment):** Assignment connects a complaint to a volunteer (**linking relationship**).
- **Admin → Volunteer/Assignment/Resource/Report:** Admin supervises system operations (**association**).
- **Report → System Entities:** Report gathers data from complaints, assignments, resources, and alerts (**dependency**).





# DOMAIN MODEL DIAGRAM

- **Victim:** Submits **Complaint/Help Request** entries (location, type, intensity). A Victim can submit multiple complaints.
- **Volunteer/NGO:** Performs relief tasks, manages **Shelter Relief Camps**, and updates **Resource/Donation** availability.
- **Donor:** Provides financial or material support recorded in the **Resource/Donation** entity.
- **Admin:** A distinct authority figure with elevated permissions. Responsibilities include approving volunteers, assigning tasks, monitoring resources, and generating **Reports**.

## Work Flow:

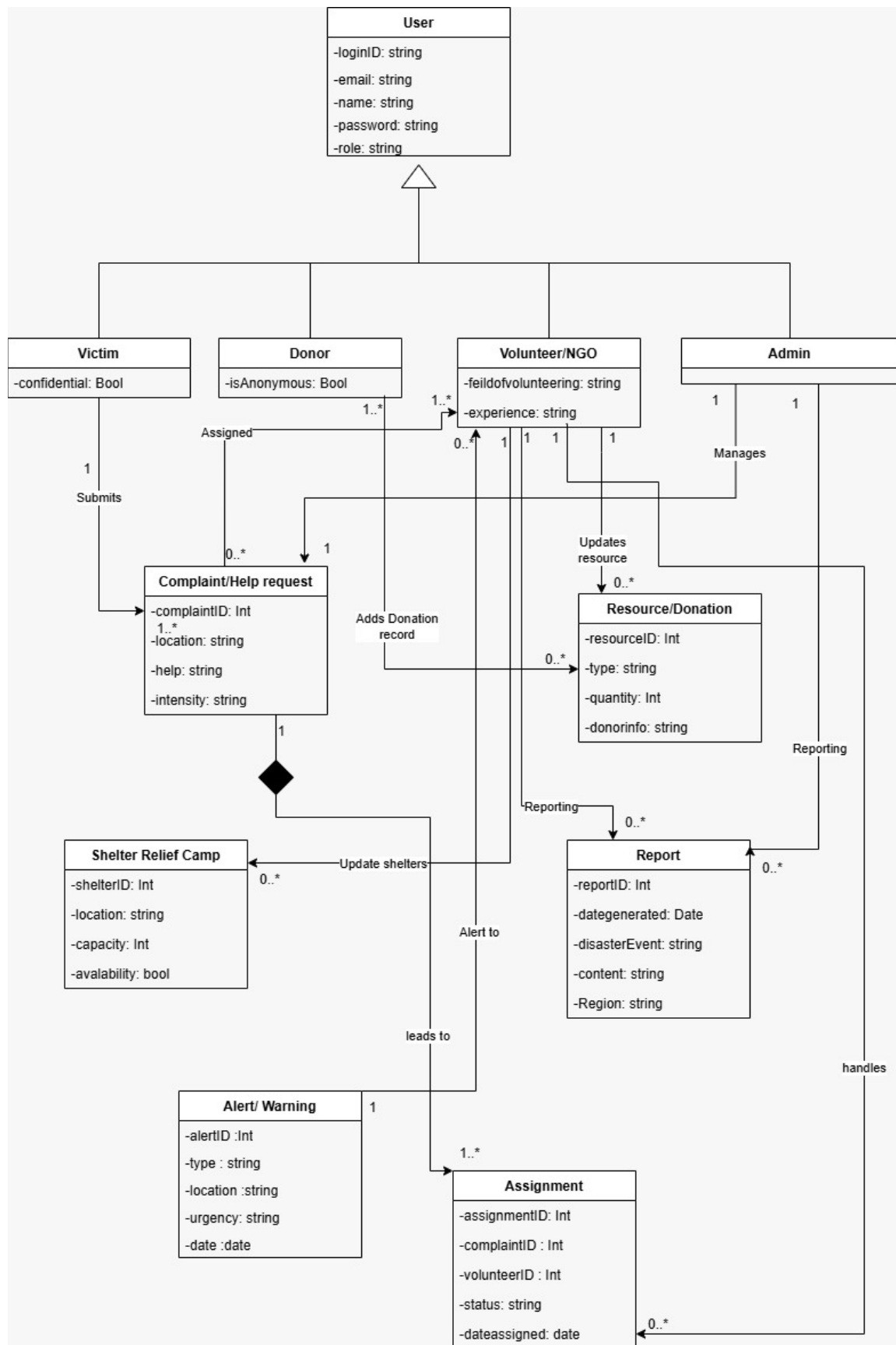
**Complaint/Help Requests** lead to one or more **Assignments**. This entity links victims' needs directly to volunteers' actions.

**Resource/Donation:** Tracks all supplies, updated by both Donors and Volunteers.

**Shelter Relief Camp:** Records temporary shelter details, updated by Volunteers.

**Alerts/Warnings:** Notifies Volunteers about urgent conditions (type, urgency, location) to facilitate immediate action.

**Report:** Provides summary information for administrative monitoring of efforts, resources, and system performance



# SEQUENCE DIAGRAM

 **Volunteer Registration:** A **Volunteer** submits details. The **System** validates the input, and the **Admin** approves the data, granting the volunteer system access.

 **Complaint Submission:** A **Victim** submits a complaint (location, type, intensity). The **System** processes the request and generates a unique ID.

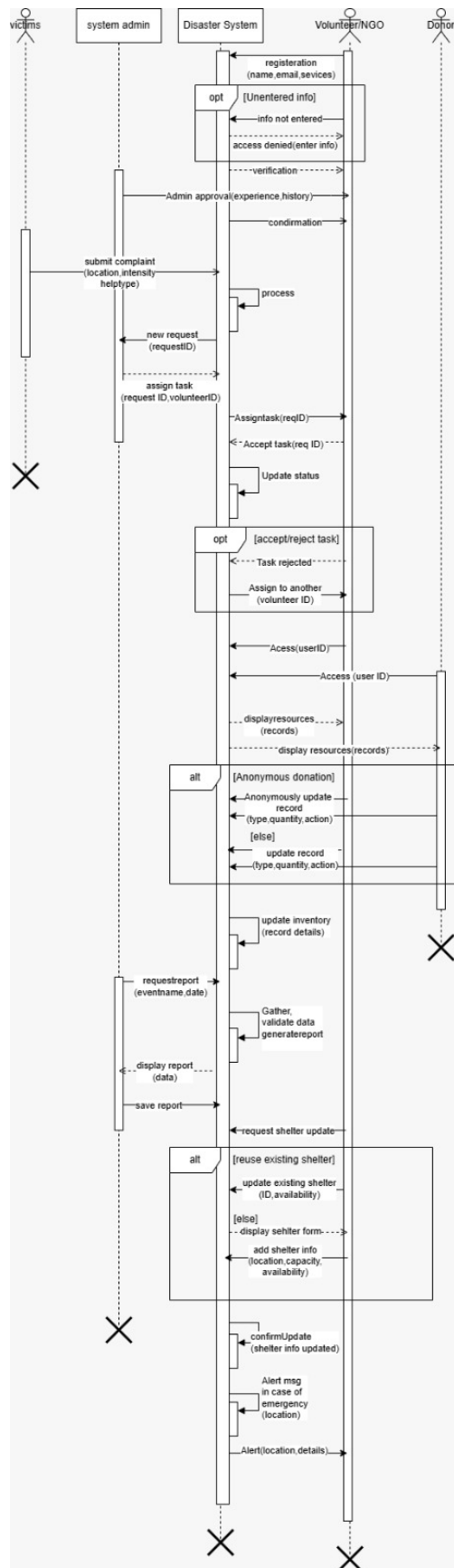
 **Task Assignment:** The **Admin/System** assigns the complaint to a **Volunteer**. If rejected, the system reassigns it. Upon acceptance, the volunteer's status is updated.

 **Resource Management:** **Volunteers, Donors, or Admins** update the resource database. The **System** records donations (type, quantity) and updates resource availability.

 **Shelter Management:** **Volunteers** update or create **Shelter** records. The **System** checks for reuse possibilities and updates location, capacity, and availability information.

 **Report Generation:** A user requests a report. The **System** gathers all necessary data, composes the report, presents it to the user, and saves it for documentation.

 **Alert/Warning Generation:** In an emergency, the **System** generates and sends an alert message (location, threat details) to **Volunteers** to enable prompt response.



## SYSTEM RISK ANALYSIS

Disaster Management and Relief System is functioning in highly critical and time-sensitive environments where even a failure, wrong data inputs, and/or any kind of delay can lead to worse consequences. Hence, there came a need for a risk analysis that could determine the risk of technology and business types facing the system.

The below tables describe a risk analysis of the system that highlights risks linked to the system and measures to take to ensure that the system is safe with effective measures for disaster response.

## TECHNICAL RISKS:

| Risk                                    | Type of risk | Probability | Consequences                                                   | Mitigation strategy                                                          |
|-----------------------------------------|--------------|-------------|----------------------------------------------------------------|------------------------------------------------------------------------------|
| System crash during peak disaster usage | Product risk | Medium      | Victims fail to submit complaint. Failure of the relief system | Load balancing must be done. High quality systems. backups must be installed |

| Risk                                    | Type of risk | Probability | Consequences                                               | Mitigation strategy                           |
|-----------------------------------------|--------------|-------------|------------------------------------------------------------|-----------------------------------------------|
| Failure of offline data synchronization | Product risk | Medium      | Volunteers do not get the complaints on time. Help delayed | Conflict resolution systems must be installed |

| Risk                                           | Type of risk | Probability | Consequences                                      | Mitigation strategy                                 |
|------------------------------------------------|--------------|-------------|---------------------------------------------------|-----------------------------------------------------|
| Database overload due to high transaction rate | Product risk | High        | Response delayed. Complaint delayed; Help delayed | Use scalable databases and performance optimization |

| Risk                                | Type of risk | Probability | Consequences                                                              | Mitigation strategy                                       |
|-------------------------------------|--------------|-------------|---------------------------------------------------------------------------|-----------------------------------------------------------|
| Incorrect disaster alert generation | Product risk | Medium      | Incorrect alerts create unnecessary panic and waste of time and resources | Alerts must be validated properly and source confirmation |

## SECURITY TYPE RISKS:

| Risk                               | Type of risk  | Probability | Consequences                                                | Mitigation strategy                                                            |
|------------------------------------|---------------|-------------|-------------------------------------------------------------|--------------------------------------------------------------------------------|
| Unauthorized access to victim data | Business risk | Medium      | The release of personal information. People will lose trust | Proper system security to protect sensitive information. End to end encryption |

| Risk                               | Type of risk | Probability   | Consequences                                                         | Mitigation strategy                                                                   |
|------------------------------------|--------------|---------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Data tampering in resource records | Product Risk | Low to Medium | Tempering must cause confusion. Problems keeping track of activities | Proper security checks every now and then. Keep track of all the logs and any changes |

| Risk                         | Type of risk  | Probability | Consequences                                        | Mitigation strategy                                                                             |
|------------------------------|---------------|-------------|-----------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Fake volunteer registrations | Business Risk | Medium      | Countless Security threats. Misuse of system access | Volunteer activities must be tracked. volunteers must be authenticated from different resources |



| Risk                              | Type of risk | Probability | Consequences                                                    | Mitigation strategy                               |
|-----------------------------------|--------------|-------------|-----------------------------------------------------------------|---------------------------------------------------|
| Weak password or credential theft | Project Risk | Medium      | Anyone can temper the whole system and mess with sensitive data | Proper password policy. Two factor authentication |

## OPERATIONAL RISKS:

| Risk                                               | Type of risk | Probability | Consequences                                             | Mitigation strategy                                                             |
|----------------------------------------------------|--------------|-------------|----------------------------------------------------------|---------------------------------------------------------------------------------|
| Volunteers becoming unavailable during emergencies | Project Risk | Medium      | Delayed response. People suffer due to insufficient help | Backup options should be available at all times. reassignment systems developed |

| Risk                                          | Type of risk  | Probability | Consequences                                                        | Mitigation strategy                                                   |
|-----------------------------------------------|---------------|-------------|---------------------------------------------------------------------|-----------------------------------------------------------------------|
| Poor coordination between NGOs and volunteers | Business Risk | Medium      | Risk of duplicate efforts. Operations will be insufficient at times | Proper coordination between the volunteers and NGO. Centralized tasks |

| Risk                                         | Type of risk | Probability | Consequences                                              | Mitigation strategy                                                                      |
|----------------------------------------------|--------------|-------------|-----------------------------------------------------------|------------------------------------------------------------------------------------------|
| Incorrect complaint information from victims | Product Risk | Medium      | A waste of time and resources. Actual victims will suffer | The location of information must be validated. Proper resources to confirm the complaint |

| Risk                                    | Type of risk | Probability | Consequences                                                                  | Mitigation strategy                                                |
|-----------------------------------------|--------------|-------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Human error while updating shelter data | Project Risk | Low         | There can be overcrowding of shelters and some shelters may remain unutilized | Data validation rules, admin verification, update history tracking |

## MANAGEMENT RISKS:

| Risk                                    | Type of risk | Probability | Consequences                                                                  | Mitigation strategy                                                |
|-----------------------------------------|--------------|-------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Human error while updating shelter data | Project Risk | Low         | There can be overcrowding of shelters and some shelters may remain unutilized | Data validation rules, admin verification, update history tracking |

| Risk                                        | Type of risk  | Probability | Consequences                                                                             | Mitigation strategy                                                    |
|---------------------------------------------|---------------|-------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Insufficient funding for system maintenance | Business Risk | Medium      | This can result in System degradation and outdated features makes the system incompetent | There must be proper Budget planning, sponsorships, phased development |

| Risk                                                | Type of risk  | Probability | Consequences                                         | Mitigation strategy                                                                 |
|-----------------------------------------------------|---------------|-------------|------------------------------------------------------|-------------------------------------------------------------------------------------|
| Legal or compliance issues related to data handling | Business Risk | Low         | There will be System shutdown, penalties due to this | The handlers of data must have proper knowledge about protection laws, legal review |

| Risk                               | Type of risk | Probability | Consequences                        | Mitigation strategy                            |
|------------------------------------|--------------|-------------|-------------------------------------|------------------------------------------------|
| Lack of long-term system ownership | Project Risk | Medium      | System abandonment after deployment | Assign responsible authority, maintenance plan |

The identified risks and methods for mitigating them reflect a proactive attitude towards the prevention of system failures and inefficiencies. Through the identification of the risks at the design and development stages of the system development life cycle, the system is anticipated to be functioning well under extreme disaster circumstances to ensure timely relief to the affected individuals and coordination between volunteers, NGOs, and donors.

# BLACK BOX TESTING

## Requirement 1:

### Volunteer registration:

- The volunteers must register themselves with their names and personal information.
- The user must add information about their role in the system.
- Every user must have their designated login ID.

| Field           | TC-01 (Valid Input)                      | TC-02 (Invalid Input)                              |
|-----------------|------------------------------------------|----------------------------------------------------|
| Date            | 10-12-25                                 | 10-12-25                                           |
| System          | Disaster management system               | Disaster management system                         |
| Objective       | Verify successful volunteer registration | Verify validation for incomplete registration      |
| Test ID         | BB-01-A                                  | BB-01-B                                            |
| Version         | 1                                        | 1                                                  |
| Test type       | Black Box Testing                        | Black Box Testing                                  |
| Input           | All required fields filled correctly     | One or more required fields left empty             |
| Expected Result | Registration successful message shown    | Error message "All required fields must be filled" |
| Actual Result   | Passed                                   | Passed                                             |

[Register](#) [Login](#) [Complaint](#) [Assign Task](#) [Resources](#) [Tracking](#) [Shelters](#) [Alerts](#) [Reports](#) [Offline](#) [View & Manage Data](#)

### Volunteer Registration

Yes

▼

NGO HELP

▼

Rescue Operation

▼

Register

Registration Successful

[1. Register](#) [2. Login](#) [3. Complaint](#) [4. Assign Task](#) [5. Resources](#) [6. Tracking](#) [7. Shelters](#) [8. Alerts](#) [9. Offline](#) [10. Reports](#) [11. Manager Panel](#)

### Volunteer Registration

Part of NGO? Yes

▼

NGO HELP

▼

Rescue Operation

▼

Please fill out this field.

Registered!

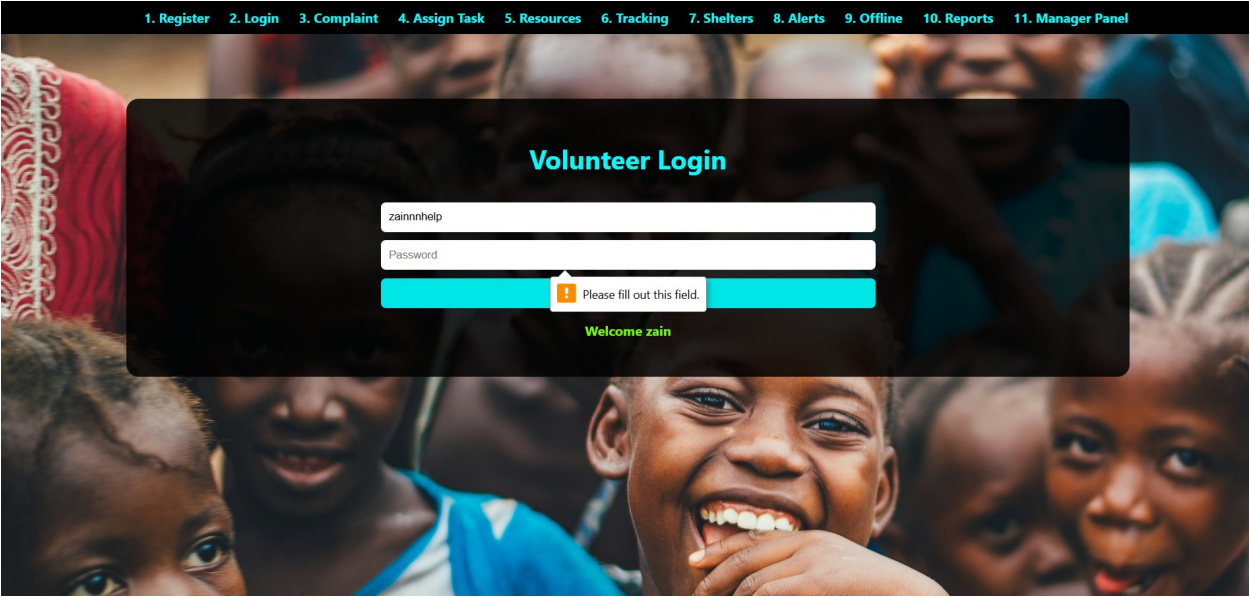
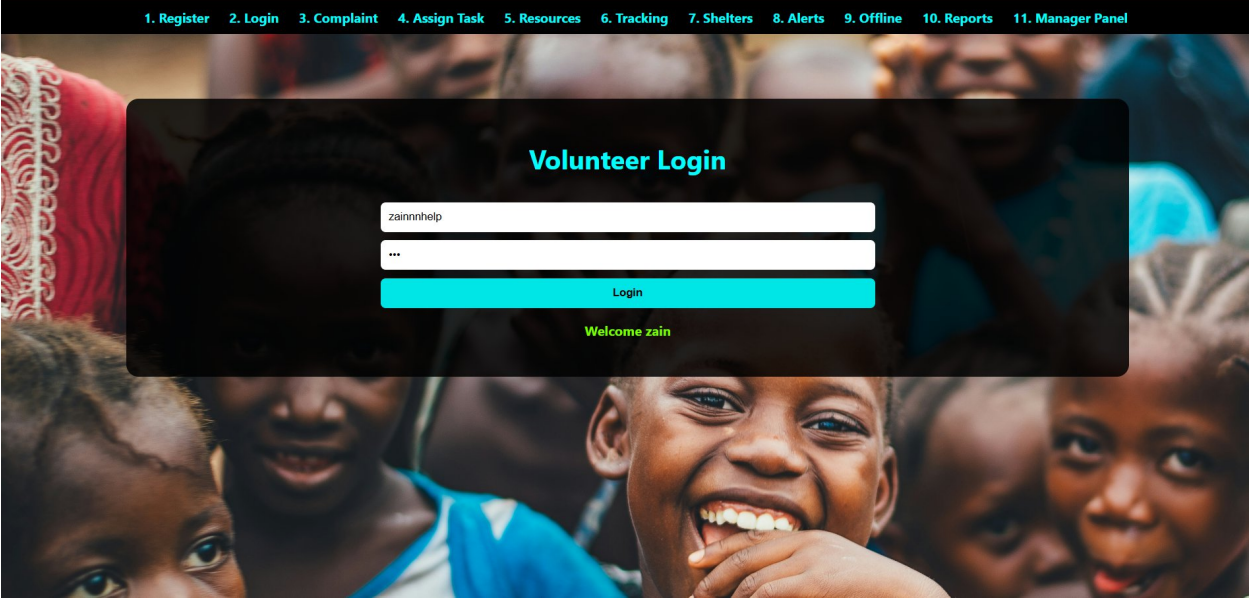
## Requirement 2:

### 2: Login page:

- Login allows users to enter username and password to access the system.
- The system prevents unauthorized access by rejecting invalid login attempts
- Error message are displayed when invalid or missing credentials are entered.

| Field           | TC-01 (Valid Input)                   | TC-02 (Invalid Input)                   |
|-----------------|---------------------------------------|-----------------------------------------|
| Date            | 10-12-25                              | 10-12-25                                |
| System          | Disaster management system            | Disaster management system              |
| Objective       | Verify login with correct credentials | Verify login with incorrect credentials |
| Test ID         | BB-02-A                               | BB-02-B                                 |
| Version         | 1                                     | 1                                       |
| Test type       | Black Box Testing                     | Black Box Testing                       |
| Input           | Correct username and password         | Incorrect username and password         |
| Expected Result | Login successful message              | Error message 'Invalid credentials'     |
| Actual Result   | Passed                                | Passed                                  |





## Requirement 3:

### 3: Complaint management center:

- The user can enter their complaints in the system complaint center.
- The victims can mention their needs and wants in case of any disaster situation.
- The system must receive and process the complaint and forward it to the NGO'S and volunteers for them to take action.

| Field           | TC-01 (Valid Input)                      | TC-02 (Invalid Input)                 |
|-----------------|------------------------------------------|---------------------------------------|
| Date            | 10-12-25                                 | 10-12-25                              |
| System          | Disaster management system               | Disaster management system            |
| Objective       | Verify complaint submission              | Verify empty complaint submission     |
| Test ID         | BB-03-A                                  | BB-03-B                               |
| Version         | 1                                        | 1                                     |
| Test type       | Black Box Testing                        | Black Box Testing                     |
| Input           | Complaint description & location entered | Empty description/location            |
| Expected Result | Complaint submitted successfully         | Complaint not submitted / error shown |
| Actual Result   | Passed                                   | Passed                                |

1. Register2. Login3. Complaint4. Assign Task5. Resources6. Tracking7. Shelters8. Alerts9. Offline10. Reports11. Manager Panel

### Victim Complaint Center

**Description of Situation**

**Needs and Wants**

**Location**

Submit Complaint

Submitted

1. Register2. Login3. Complaint4. Assign Task5. Resources6. Tracking7. Shelters8. Alerts9. Offline10. Reports11. Manager Panel

### Victim Complaint Center

**Description of Situation**

**Needs and Wants**

**Location**

Submit Complaint

⚠ Invalid Fields: Required!

## Requirement 4:

### 4: Assigning tasks to volunteers:

- The complaint of the victim is processed by the system.
- The system shows the complaint on the user's side.
- The system assigns the complaint to the available volunteers by tracking the volunteer activity.
- The volunteers who are assigned the task then take action to resolve the issue.

| Field           | TC-01 (Valid Input)                   | TC-02 (Invalid Input)                |
|-----------------|---------------------------------------|--------------------------------------|
| Date            | 10-12-25                              | 10-12-25                             |
| System          | Disaster management system            | Disaster management system           |
| Objective       | Assign task to volunteer              | Validate missing task info           |
| Test ID         | BB-04-A                               | BB-04-B                              |
| Version         | 1                                     | 1                                    |
| Test type       | Black Box Testing                     | Black Box Testing                    |
| Input           | Volunteer selected + task description | Volunteer not selected or task empty |
| Expected Result | Task assigned successfully            | Error message displayed              |
| Actual Result   | Passed                                | Passed                               |



1. Register2. Login3. Complaint4. Assign Task5. Resources6. Tracking7. Shelters8. Alerts9. Offline10. Reports11. Manager Panel

### Assign Task

Select Volunteer

Task Description

Task Location

Assign Task

✔

Task Assigned Successfully! Volunteer has been notified.

1. Register2. Login3. Complaint4. Assign Task5. Resources6. Tracking7. Shelters8. Alerts9. Offline10. Reports11. Manager Panel

### Assign Task

zain (Available)

he will save people from drowning

Task Location

Assign Task

⚠

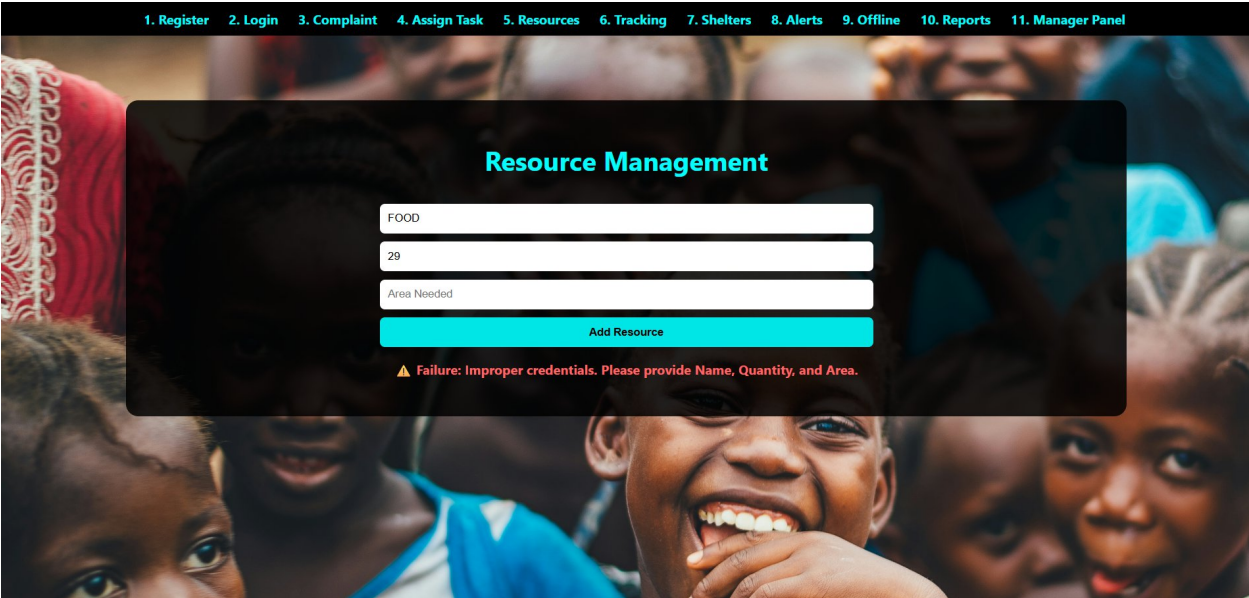
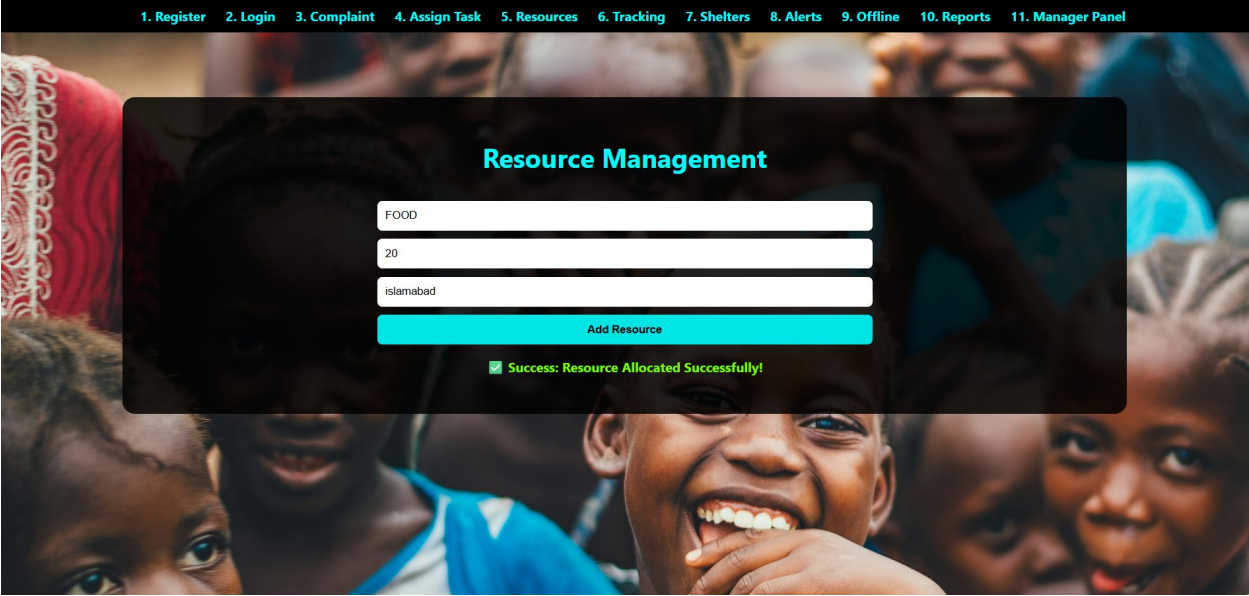
Assignment Failed: Invalid Entry. Please select a volunteer and fill all task details.

# Requirement 5:

## 5: Management of resources:

- The system keeps track of the resources that are available for the victims.
- The authorized personnel can also alter the records by entering or deleting any extra information.
- They can enter the any updates in the donation about any new donations.

| Field           | TC-01 (Valid Input)         | TC-02 (Invalid Input)          |
|-----------------|-----------------------------|--------------------------------|
| Date            | 10-12-25                    | 10-12-25                       |
| System          | Disaster management system  | Disaster management system     |
| Objective       | Add new resource            | Validate empty resource fields |
| Test ID         | BB-05-A                     | BB-05-B                        |
| Version         | 1                           | 1                              |
| Test type       | Black Box Testing           | Black Box Testing              |
| Input           | Resource name & quantity    | Empty name or quantity         |
| Expected Result | Resource added successfully | Resource not added             |
| Actual Result   | Passed                      | Passed                         |



# Requirement 6:

## 6: Track of volunteer activities:

- The system keeps track of the activities of different volunteers.
- It checks if the volunteers are available.
- It shows what volunteers are assigned to what task at what time.
- The volunteers can add and update their status according to their availability.

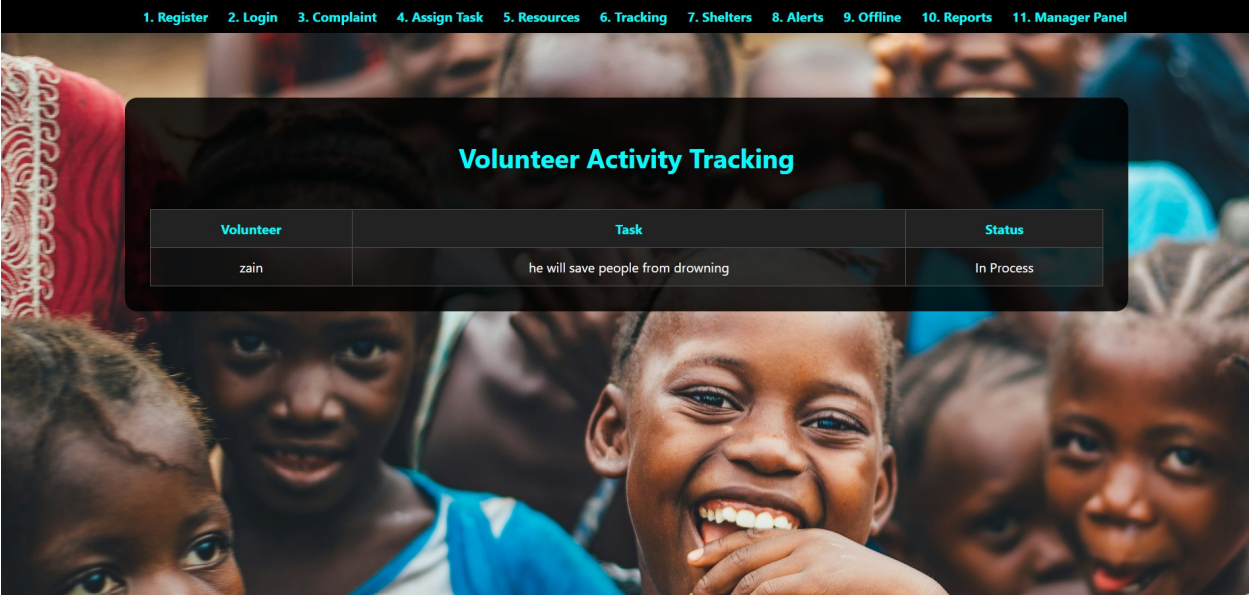
| Field           | TC-01 (Valid Input)        | TC-02 (Invalid Input)      |
|-----------------|----------------------------|----------------------------|
| Date            | 10-12-25                   | 10-12-25                   |
| System          | Disaster management system | Disaster management system |
| Objective       | Track assigned tasks       | No tasks assigned          |
| Test ID         | BB-06-A                    | BB-06-B                    |
| Version         | 1                          | 1                          |
| Test type       | Black Box Testing          | Black Box Testing          |
| Input           | Assigned task exists       | No task data               |
| Expected Result | Tasks displayed correctly  | Empty tracking table       |
| Actual Result   | Passed                     | Passed                     |



1. Register2. Login3. Complaint4. Assign Task5. Resources6. Tracking7. Shelters8. Alerts9. Offline10. Reports11. Manager Panel

Volunteer Activity Tracking

| Volunteer | Task                              | Status     |
|-----------|-----------------------------------|------------|
| zain      | he will save people from drowning | In Process |

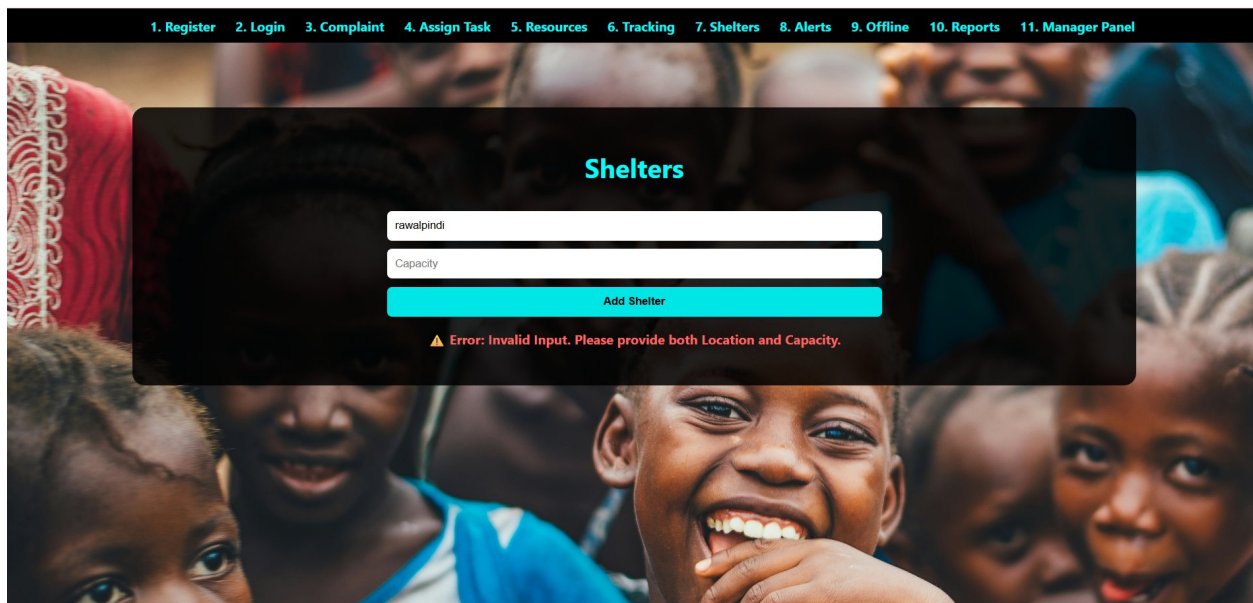
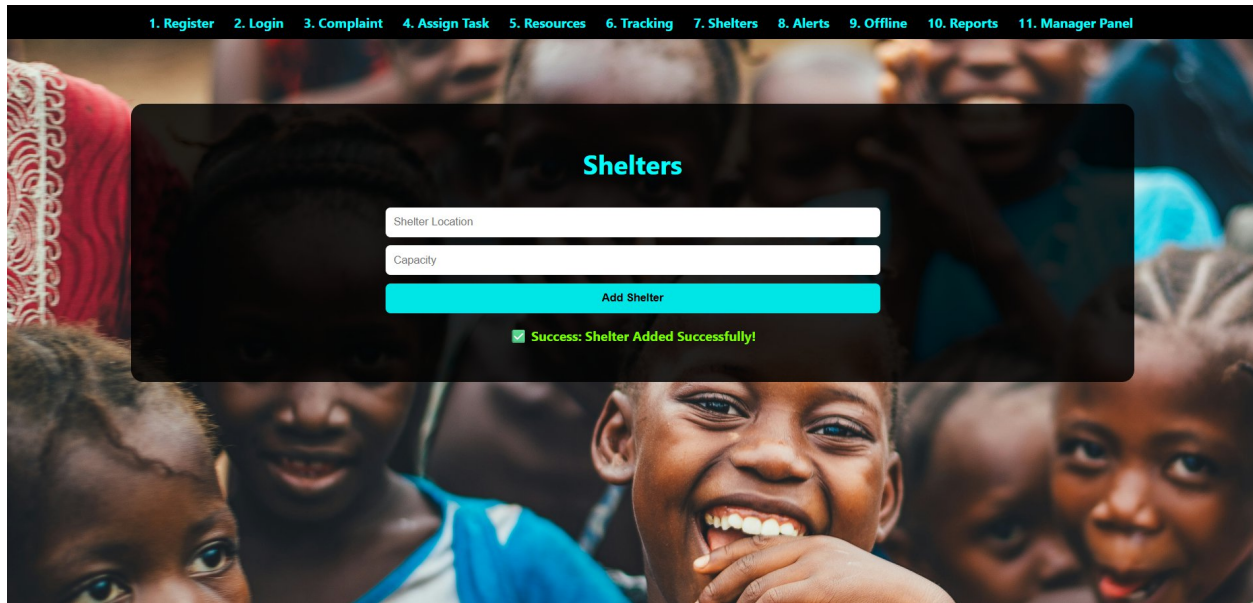


# Requirement 7:

## 7: Checking shelters and relief camps:

- The system keeps a track of the availability of shelter camps for the victims in case of a disaster.
- The volunteers can also update the availability or unavailability of the shelters.
- The system can also keep track of the location of the relief camps
- It can also track the amount of people a relief camp can accommodate.

| Field           | TC-01 (Valid Input)        | TC-02 (Invalid Input)         |
|-----------------|----------------------------|-------------------------------|
| Date            | 10-12-25                   | 10-12-25                      |
| System          | Disaster management system | Disaster management system    |
| Objective       | Add shelter details        | Validate missing shelter info |
| Test ID         | BB-07-A                    | BB-07-B                       |
| Version         | 1                          | 1                             |
| Test type       | Black Box Testing          | Black Box Testing             |
| Input           | Location & capacity        | Empty location or capacity    |
| Expected Result | Shelter added              | Shelter not added             |
| Actual Result   | Passed                     | Passed                        |

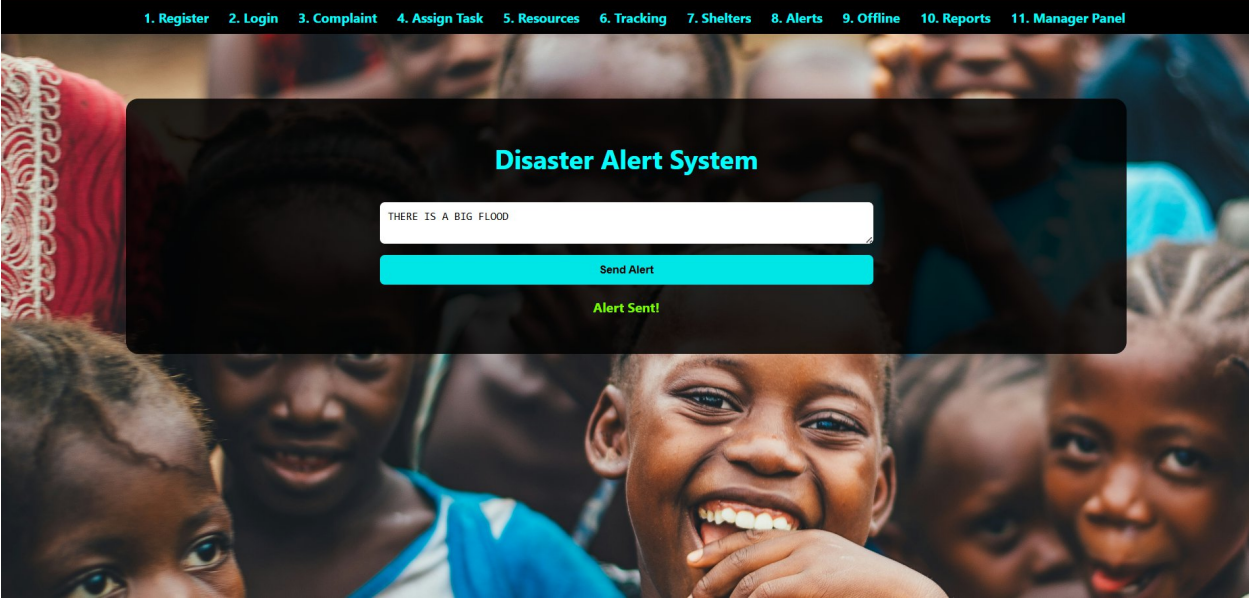


## Requirement 8:

### 8: Disaster alert system:

- The system helps the volunteers can get a warning message or an alert in case of any kind of emergency situation.
- This will help the volunteers to take an immediate action upon the situation and prepare necessary reliefs for the affected people.
- The alerts are traced so the system can inform the ngo/volunteers about the location.

| Field           | TC-01 (Valid Input)        | TC-02 (Invalid Input)      |
|-----------------|----------------------------|----------------------------|
| Date            | 10-12-25                   | 10-12-25                   |
| System          | Disaster management system | Disaster management system |
| Objective       | Send disaster alert        | Send empty alert           |
| Test ID         | BB-08-A                    | BB-08-B                    |
| Version         | 1                          | 1                          |
| Test type       | Black Box Testing          | Black Box Testing          |
| Input           | Alert message entered      | No alert message           |
| Expected Result | Alert sent to volunteers   | Alert not sent             |
| Actual Result   | Passed                     | Passed                     |



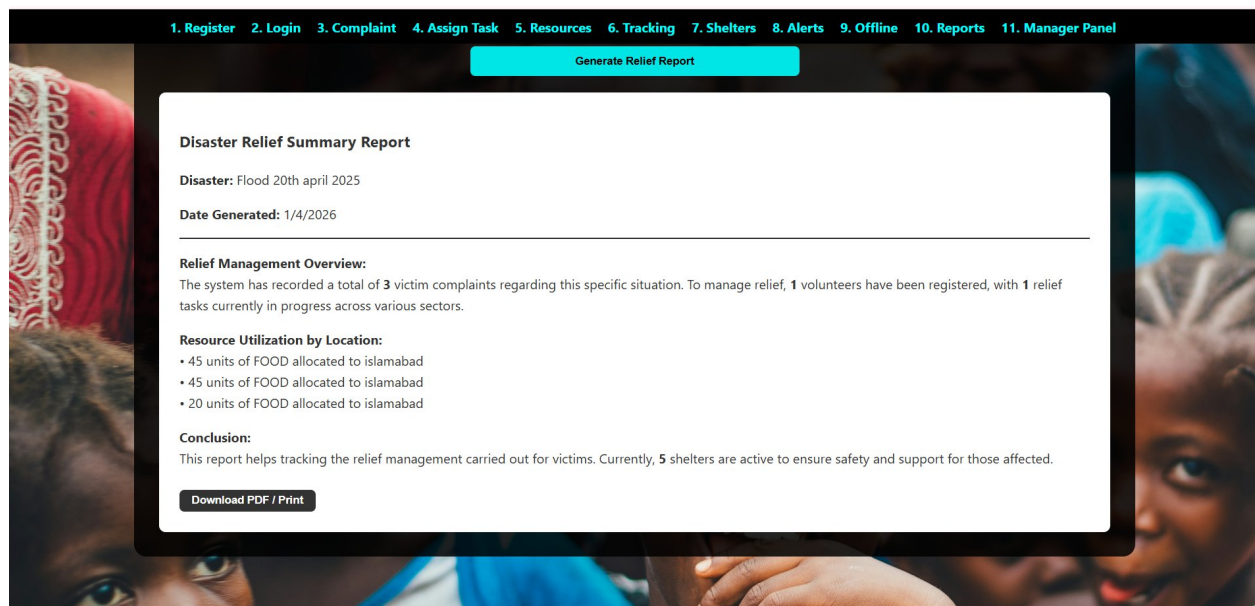
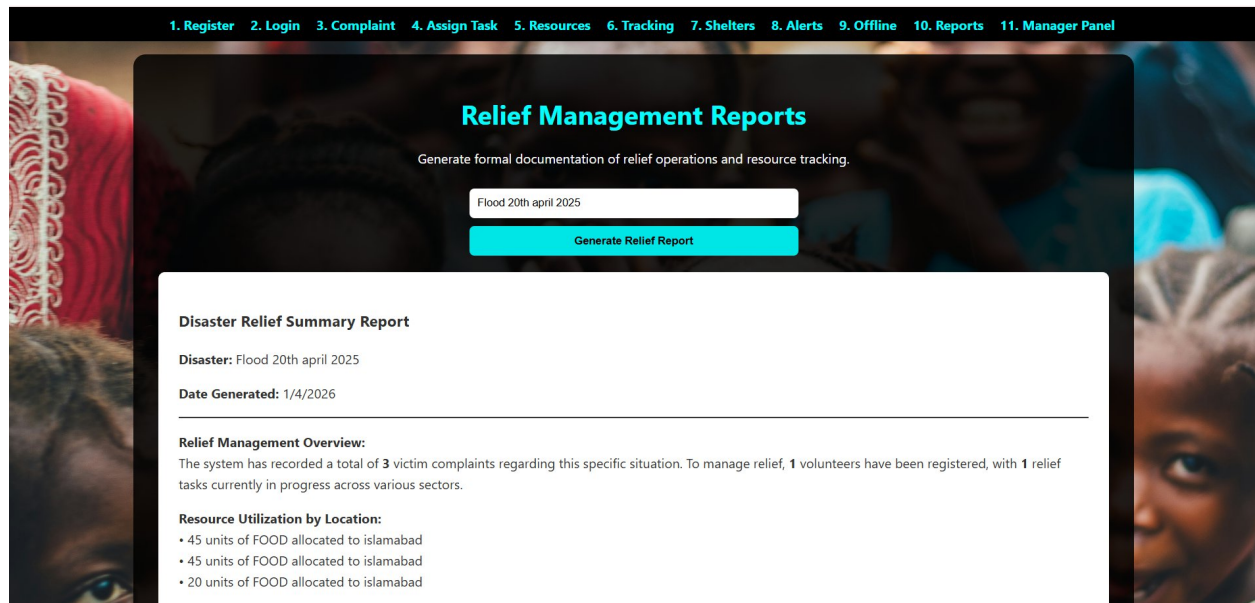
# Requirement 9:

## 9: Generation of reports:

- The system is able generated well written reports for the user.
- These reports may contain information about a specific disaster.
- These reports can be about the number of resources used in a specific situation or at a specific location.
- These reports help the users to keep track of the relief management they carried out for the victims.

| Field           | TC-01 (Valid Input)                                                                          | TC-02 (Invalid Input)                                        |
|-----------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Date            | 10-12-25                                                                                     | 10-12-25                                                     |
| System          | Disaster management system                                                                   | Disaster management system                                   |
| Objective       | Verify report generation when data exists                                                    | Verify report generation when no data exists                 |
| Test ID         | BB-09-A                                                                                      | BB-09-B                                                      |
| Version         | 1                                                                                            | 1                                                            |
| Test type       | Black Box Testing                                                                            | Black Box Testing                                            |
| Input           | User clicks "Generate Summary" button                                                        | User clicks "Generate Summary" with no records stored        |
| Expected Result | System generates summary showing number of users, complaints, resources, shelters, and tasks | System generates report with zero values instead of crashing |
| Actual Result   | Passed                                                                                       | Passed                                                       |





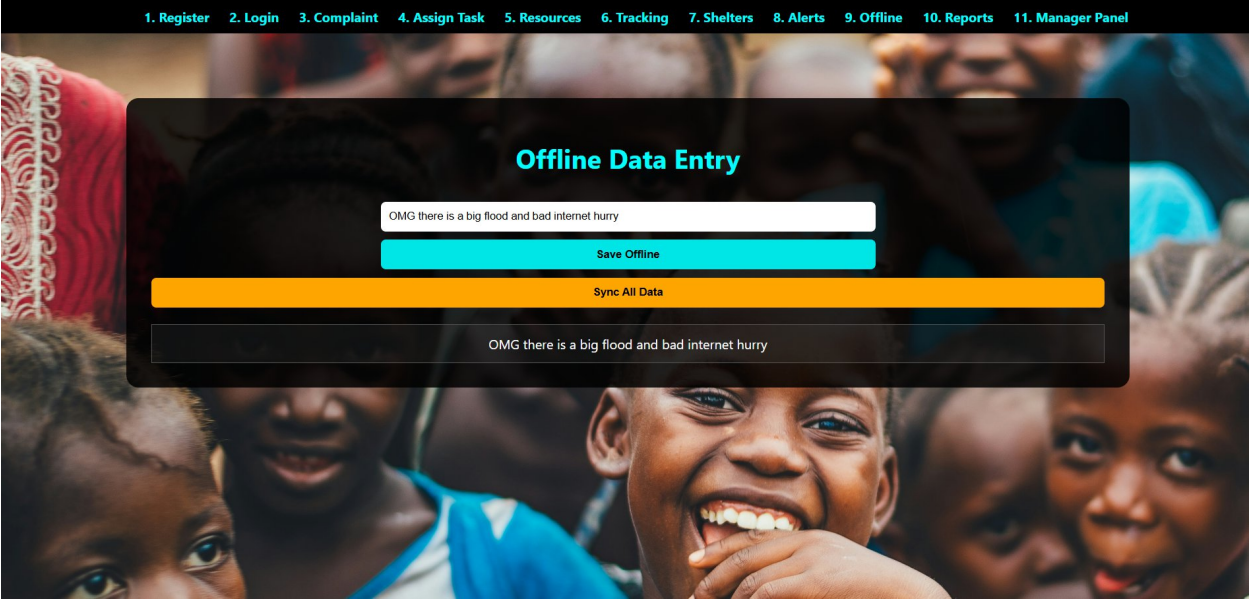
# Requirement 10:

## 10: Handling system without internet:

- This system helps manage and helps enter the information in the system without the need of internet.
- The system and the information will synchronize as soon as the system catches any signal of internet.

| Field           | TC-01 (Valid Input)        | TC-02 (Invalid Input)        |
|-----------------|----------------------------|------------------------------|
| Date            | 10-12-25                   | 10-12-25                     |
| System          | Disaster management system | Disaster management system   |
| Objective       | Save offline data          | Validate empty offline input |
| Test ID         | BB-10-A                    | BB-10-B                      |
| Version         | 1                          | 1                            |
| Test type       | Black Box Testing          | Black Box Testing            |
| Input           | Offline complaint entered  | Empty input                  |
| Expected Result | Data saved offline         | Error message shown          |
| Actual Result   | Passed                     | Passed                       |





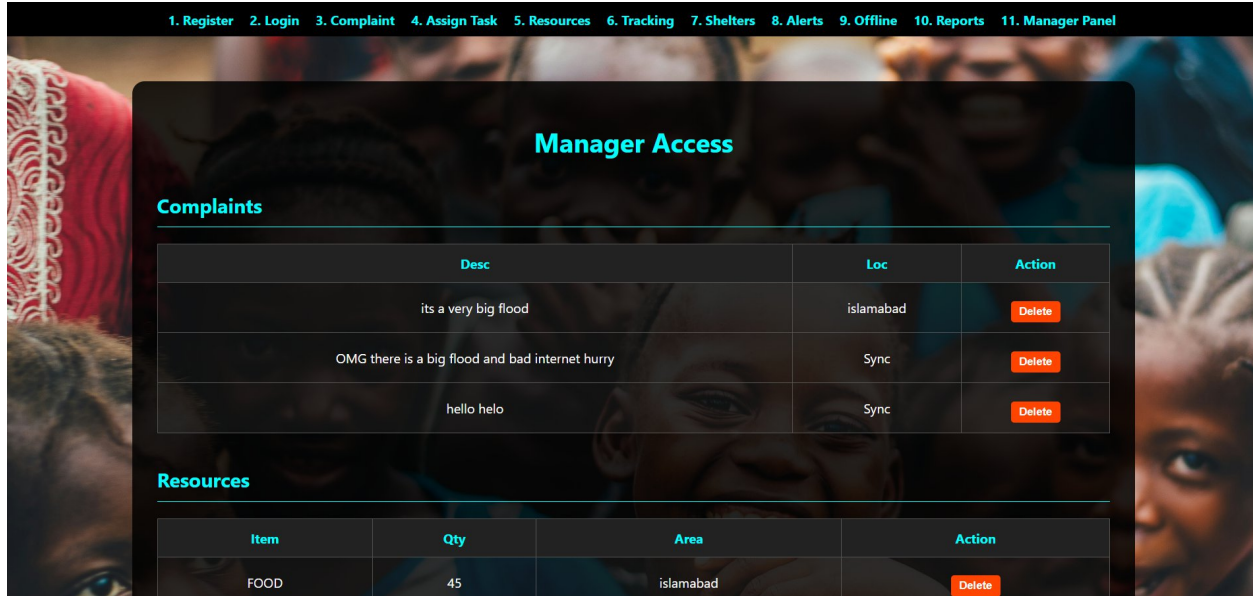
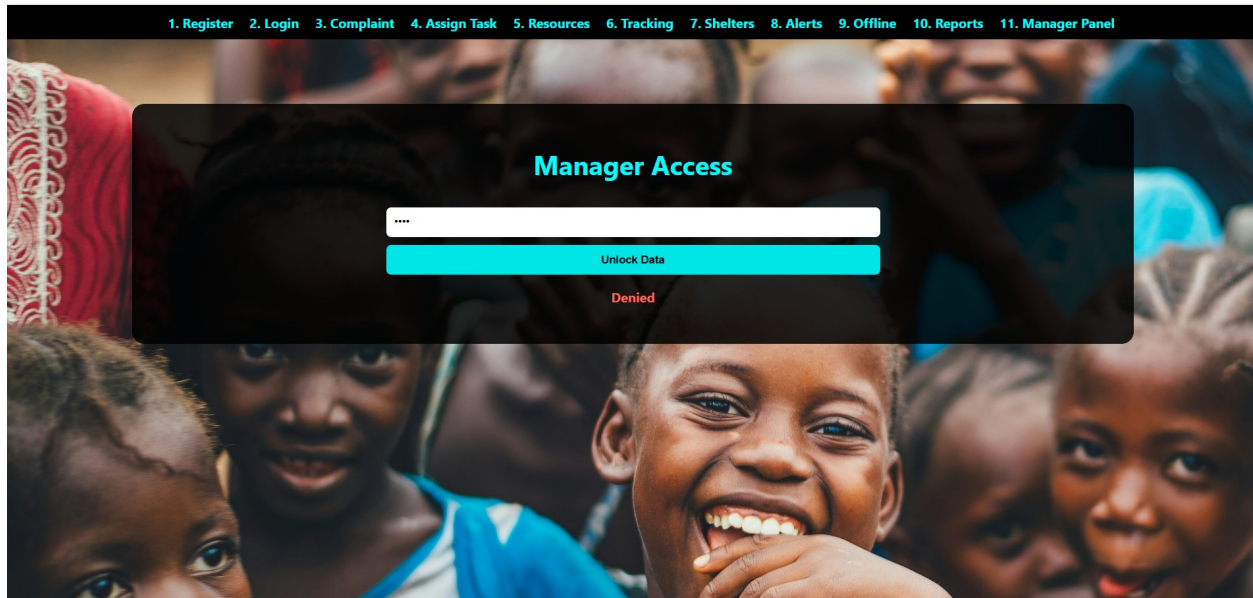
# Requirement 11:

## 11: Security:

- The system must provide full confidentiality to the victims and members.
- The donors must not be able to access any information about the victims.
- Only the managers must be able to view everything about the victims, donors, NGO's etc.

| Field           | TC-01 (Valid Input)          | TC-02 (Invalid Input)                  |
|-----------------|------------------------------|----------------------------------------|
| Date            | 10-12-25                     | 10-12-25                               |
| System          | Disaster management system   | Disaster management system             |
| Objective       | Restrict unauthorized access | Prevent donor from viewing victim data |
| Test ID         | BB-11-A                      | BB-11-B                                |
| Version         | 1                            | 1                                      |
| Test type       | Black Box Testing            | Black Box Testing                      |
| Input           | Authorized user access       | Unauthorized user access               |
| Expected Result | Data visible                 | Access denied                          |
| Actual Result   | Passed                       | Passed                                 |

Password was admin123 but we entered 222:



1. Register2. Login3. Complaint4. Assign Task5. Resources6. Tracking7. Shelters8. Alerts9. Offline10. Reports11. Manager Panel

Resources

| Item | Qty | Area      | Action |
|------|-----|-----------|--------|
| FOOD | 45  | islamabad | Delete |
| FOOD | 20  | islamabad | Delete |

Shelters

| Loc                     | Cap | Action |
|-------------------------|-----|--------|
| islamabad               | 20  | Delete |
| Islamabad near the park | 20  | Delete |

Tasks

| Vol  | Status     | Reason                           | Action |
|------|------------|----------------------------------|--------|
| zain | In Process | he can save people from drowning | Delete |

Users

| Name | Field   | Status | Action |
|------|---------|--------|--------|
| zain | General | Busy   | Delete |

Lock Records